

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and trained **deep learning networks** for object detection and classification tasks, processing terabytes of automotive sensor data to improve scene understanding by **30%**.
- Implemented **Python** scripts to automate data acquisition and preprocessing pipelines, ensuring data integrity and reducing manual data handling time by **40%**.
- Refactored **SQL queries** across large PostgreSQL datasets, optimizing data retrieval for statistical analysis and reducing query execution times by an average of **25%**.
- Contributed to the development of **Java Spring Boot** REST APIs for inter-service communication, enhancing system integration and supporting new feature deployments.
- Managed production incidents for enterprise applications, maintaining **99.95% SLA** adherence through proactive monitoring with **Splunk** and rapid issue resolution.
- Applied machine learning fundamentals from MSc coursework to analyze experimental test results, performing **statistical analysis** and fine-tuning models for improved performance metrics.

SKILLS

Deep Learning – Keras, PyTorch, scikit-learn, TensorFlow

Computer Vision – OpenCV, 3D object detection, classification, structure-from-motion, localisation & mapping

Programming Languages – Python, Java, R, English

Databases & Data Processing – SQL, PostgreSQL, MySQL, Oracle, Power Query, DAX, data modelling

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux, Git, Jenkins

Data Science & Analytics – pandas, NumPy, RStudio, statistical analysis, machine learning, data warehousing, BI dashboards

Software Architecture – REST APIs, data structures

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Automated Driving Scene Understanding Platform

- Developed a **Python**-based data pipeline to ingest and preprocess diverse automotive sensor data (camera, lidar) for training deep learning models.
- Implemented **PyTorch** for training Convolutional Neural Networks (CNNs) focused on 3D object detection and trajectory mapping, achieving **92% accuracy** on benchmark datasets.

- Integrated OpenCV for real-time image processing and feature extraction, supporting the development of collision mitigation algorithms.

Real-Time Vehicle Environment Mapping System

- Designed and implemented a **Java Spring Boot** microservice to fuse data from multiple perception modules, creating a comprehensive real-time map of the vehicle's surroundings.
- Managed data flow and storage using **PostgreSQL**, ensuring efficient retrieval for downstream analysis and decision-making processes.
- Deployed containerized services using **Docker** on AWS EC2 instances, optimizing resource utilization and enabling scalable deployment for fleet testing.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)